

Practical No: 05

Aim: To delete documents from a MongoDB collection.

To understand and perform document deletion operations in MongoDB using `deleteOne()` and `deleteMany()` methods, and to learn how MongoDB removes specific or multiple documents from a collection based on filter conditions.

Theory:

Deletion of documents is an essential database operation that allows the user to remove unwanted, outdated, or irrelevant data from a collection. In MongoDB, deletion operations are performed using methods such as **`deleteOne()`** and **`deleteMany()`**, which accept a filter condition to identify the documents to be removed. MongoDB does not delete an entire collection unless explicitly commanded; instead, it removes only those documents that match the provided criteria.

The **`deleteOne()`** method removes the **first document** that matches the filter. If multiple documents satisfy the condition, only the first encountered document is deleted, making the operation predictable and controlled. For example, deleting a student with a specific name ensures only one matching record is removed. MongoDB returns an acknowledgement containing details such as `acknowledged: true` and `deletedCount: 1`, indicating the number of documents deleted.

The **`deleteMany()`** method is used when all documents matching a particular filter need to be deleted. This is useful in scenarios where multiple records belong to the same category, department, or status. For example, deleting all students from the ECE department removes every document where `dept: "ECE"`. This method is efficient and avoids the need for repeated deletion commands.

Commands:

```
db.students.deleteOne({ name: "Rahul" })
db.students.deleteMany({ dept: "ECE" })
```

Expected Output:

Matching documents removed.

Viva Questions:

1. What is document deletion in MongoDB?

Document deletion in MongoDB is the process of removing one or more documents from a collection based on a specified condition.

2. Which methods are used to delete documents in MongoDB?

MongoDB uses `deleteOne()` and `deleteMany()` methods to delete documents from a collection.

3. What is the difference between `deleteOne()` and `deleteMany()`?

`deleteOne()` removes only the first matching document, whereas `deleteMany()` removes all documents that match the given filter condition.

Conclusion:

This practical demonstrated the use of `deleteOne()` and `deleteMany()` methods to remove documents from a MongoDB collection. The experiment helped in understanding how MongoDB performs controlled and bulk deletions using filter conditions. Proper knowledge of deletion operations is essential for efficient database management, data cleanup, and maintaining data integrity in real-world applications.